

torch.linalg.cross#

<https://pytorch.org/docs/stable/generated/torch.linalg.cross.html>

Description:

Computes the cross product of two 3-dimensional vectors. Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of vectors, for which it computes the product along the dimension dim. It broadcasts over the batch dimensions. input (Tensor) – the first input tensor. other (Tensor) – the second input tensor. dim (int, optional) – the dimension along which to take the cross-product. Default: -1.

Function Signature

```
torch.linalg.cross(input, other, *, dim=-1, out=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor. Ignored if None. Default: None.

Code Examples

```
>>> a = torch.randn(4, 3)
>>> a
tensor([[ -0.3956,  1.1455,  1.6895],
        [ -0.5849,  1.3672,  0.3599],
        [ -1.1626,  0.7180, -0.0521],
        [ -0.1339,  0.9902, -2.0225]])
>>> b = torch.randn(4, 3)
>>> b
tensor([[ -0.0257, -1.4725, -1.2251],
        [ -1.1479, -0.7005, -1.9757],
        [ -1.3904,  0.3726, -1.1836],
        [ -0.9688, -0.7153,  0.2159]])
>>> torch.linalg.cross(a, b)
tensor([[ 1.0844, -0.5281,  0.6120],
        [-2.4490, -1.5687,  1.9792],
        [-0.8304, -1.3037,  0.5650],
        [-1.2329,  1.9883,  1.0551]])
>>> a = torch.randn(1, 3) # a is broadcast to match shape of b
>>> a
tensor([[ -0.9941, -0.5132,  0.5681]])
>>> torch.linalg.cross(a, b)
tensor([[ 1.4653, -1.2325,  1.4507],
        [ 1.4119, -2.6163,  0.1073],
        [ 0.3957, -1.9666, -1.0840],
        [ 0.2956, -0.3357,  0.2139]])
```

Detailed Documentation

Documentation

Computes the cross product of two 3-dimensional vectors.

Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of vectors, for which it computes the product along the dimension dim. It broadcasts over the batch dimensions.

input (Tensor) – the first input tensor.

other (Tensor) – the second input tensor.

dim (int, optional) – the dimension along which to take the cross-product. Default: -1.

out (Tensor, optional) – the output tensor. Ignored if None. Default: None.